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SCHOOL OF APPLIED SCIENCE

PROJECT SYNOPSIS REPORT

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PROJECT TITLE:	FACIAL RECOGNITION SYSTEM FOR MISSING PERSON IDENTIFICATION
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INTRODUCTION

In today's rapidly growing population, cases of missing persons especially children have become a serious social concern. Traditional methods of identifying missing individuals rely heavily on manual records, posters, and human memory, which are often slow, inefficient, and prone to errors. With the advancement of Artificial Intelligence and Computer Vision, automated identification systems can significantly improve the speed and accuracy of such processes.

ReuniteAI is an intelligent software system designed to assist in identifying missing persons by using facial recognition technology. The system allows users to upload images of unidentified individuals and compares them against a database of previously reported missing persons. By leveraging Python-based machine learning libraries and image processing techniques, ReuniteAI aims to provide a reliable and scalable solution to reconnect missing individuals with their families.

The project emphasizes the practical application of Python and its core libraries, NoSQL databases, and web technologies to solve a real-world social problem. It also demonstrates how modern software systems can integrate AI-driven decision-making with user-friendly interfaces.

OBJECTIVE

The primary objective of **ReuniteAI** is to design and implement an intelligent system that accurately identifies missing persons using facial recognition techniques.

Key Objectives:

1. To develop an AI-based facial recognition system that can analyse and compare human facial features efficiently.
2. To automate the missing person identification process, reducing dependency on manual searches.
3. To provide a secure and user-friendly platform for reporting and searching missing individuals.
4. To store and manage missing person data effectively using a NoSQL database.

5. To assist families, authorities, and social organizations by enabling faster and more reliable identification.

SCOPE OF THE PROJECT

The scope of ReuniteAI is focused on building a functional, scalable, and intelligent identification system with real-world applicability.

Project Scope Includes:

- Image-based face detection and recognition.
- Secure user authentication and authorization.
- Admin-level access for managing users and records.
- Storage of facial data and personal details in a NoSQL database.
- Matching uploaded images with existing missing person records.
- Adding new missing cases when no match is found.

Project Limitations:

- Accuracy depends on image quality and lighting conditions.
- Real-time CCTV integration is not included in the current scope.
- The system works best with frontal facial images.

Despite these limitations, the project provides a strong foundation for future enhancements such as real-time surveillance integration and deep learning optimization.

MODULES OF THE PROJECT

The ReuniteAI system is divided into well-defined modules to ensure modularity, scalability, and ease of maintenance.

1. User Module

- User registration and login.

- Uploading images of unidentified individuals.
- Viewing match results and contact details (if a match is found).
- Submitting new missing person reports when no match exists.

2. Admin Module

- Admin authentication.
- Viewing and managing all users.
- Managing missing person records.
- Approving, updating, or removing cases from the database.

3. Face Recognition Module

- Face detection using computer vision techniques.
- Feature extraction and encoding of facial landmarks.
- Comparison of uploaded faces with stored encodings.
- Match confidence calculation.

4. Database Management Module

- Storage of user data and missing person records.
- Efficient retrieval and updating of facial data.
- Handling unstructured and semi-structured data using NoSQL.

TECHNOLOGIES USING

The project uses modern, industry-relevant technologies to ensure efficiency and scalability.

Technology	Purpose
Python	The core programming language for the application logic and AI scripts.

Flask	The lightweight web framework used to handle backend API requests and serve the application
Numpy	Used for high-performance numerical calculations and handling image data arrays.
OpenCV	Used for real-time computer vision tasks, such as reading video streams and processing images.
face_recognition	The library responsible for detecting faces and generating facial encodings for matching.
MongoDB	The NoSQL database used to store flexible data structures like user profiles and missing person reports.
HTML, CSS, JavaScript	The frontend technologies used to create the user interface in the browser.

EXPECTED OUTPUT

The successful implementation of ReuniteAI is expected to deliver the following outcomes:

1. A fully functional missing person identification system using facial recognition.
2. Accurate and efficient face matching results with minimal processing time.
3. A secure and structured database for storing sensitive personal data.
4. Improved chances of reuniting missing individuals with their families.
5. A scalable foundation for future AI-driven enhancements.

The project demonstrates how Artificial Intelligence and software engineering principles can be applied to solve socially impactful problems, making ReuniteAI both technically strong and socially relevant.